# Musavvir Haque

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## Research Profile

Early-career researcher in **computational materials science**, with hands-on experience in large-scale atomistic simulation, first-principles (DFT) electronic-structure analysis, and machine-learning surrogate modelling. Experience running and automating high-throughput simulation workflows on HPC resources, and using data-driven models to speed up materials property and defect-stability prediction. Applications include fusion-relevant materials, two-dimensional materials, and semiconductors, alongside earlier work in signal processing and embedded systems. Two manuscripts under review (one first-authored) and three conference papers.

## Research Interests

Computational Materials Science · Density Functional Theory & Electronic Structure · Molecular Dynamics & Multiscale Simulation · Machine Learning for Materials Discovery · Defect Physics & Radiation Damage · Two-Dimensional & Functional Materials · Plasma–Material Interactions · High-Performance Computing

## Education

**B.Sc. (Engg.) in Electrical and Electronic Engineering** **CGPA: 3.47 / 4.00**
Jashore University of Science and Technology (JUST), Bangladesh *2022 – 2026*

- Relevant coursework: Electrical Engineering Materials, Electromagnetic Theory, Nuclear Engineering, Plasma Engineering, VLSI Design and Technology, Digital Signal Processing, Numerical Methods, Control Systems
- Class rank: Top 10 in the batch
- Thesis (Grade: A+): Hydrogen Isotope Retention and Oxide Dynamics in Tungsten with Plasma Exposure for Fusion Reactor Applications

**Higher Secondary Certificate (Science)** **GPA: 5.00 / 5.00**
Notre Dame College, Dhaka, Bangladesh *2018 – 2021*

## Publications

***Journal Articles (Under Review)***

**[J1] Musavvir Haque**, Abdullah Shadek Fahim, Saklain Mustak, Nure Modina Sathi, Jahedul Islam, Md. Tariquzzaman, and Md. Amzad Hossain, "Hydrogen Implantation and Diffusion in BCC Tungsten: A Systematic Molecular Dynamics Study over 5–150 eV and 300–1500 K," *Fusion Engineering and Design*, Under Review (MS# FUSENGDES-D-26-00360R1, Major Revision Submitted Sep. 2026).

*800 classical MD simulations in LAMMPS; resolves non-monotonic H retention, Arrhenius diffusion parameters, and confirms the absence of displacement damage across an 8×10 energy–temperature parameter space in BCC tungsten.*

**[J2]** A. S. Fahim, **Musavvir Haque**, A. Taher, N. M. Sathi, J. Islam, Md. M. Hasan, and Md. Amzad Hossain, "A Gatekeeper-Constrained Dual-Stage Random Forest Surrogate for Multi-Output Ion Implantation Prediction Across Si, SiC, and GaAs," *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms*, Under Review (MS# W2-0665, Status Updated Sep. 2026). Preprint: doi:10.2139/ssrn.7580999

***B.Sc. Thesis***

**[T1] Hydrogen Isotope Retention and Oxide Dynamics in Tungsten with Plasma Exposure for Fusion Reactor Applications**

*B.Sc. Thesis, Jashore University of Science and Technology* *Supervisor: Md. Tariquzzaman, Asst. Prof.*

Multiscale framework integrating DFT/MD-derived physical trends with ML for hydrogen isotope retention and oxide-layer evolution in plasma-exposed tungsten. Synthetic datasets emulate experimentally realistic plasma–material interaction behaviour. **XGBoost**: MAE = 0.017 eV, $R^2 = 0.88$ (binding energy); MAE = 0.91%, $R^2 = 0.87$ (crack density). Validated against TDS experiments and WEST tokamak observations.

*Conference Papers*

**[C1] Musavvir Haque** *et al.*, "Secure Academic Assessment Using IoT Based Dual Mode Biometric Attendance System for University," *2025 Int. Conf. on Quantum Photonics, AI, and Networking (QPAIN)*, Rangpur, Jul. 2025. doi:10.1109/QPAIN66474.2025.11171806

**[C2] Musavvir Haque**, R. I. Rony, S. Mustak and I. Khalil, "Adaptive Multi-Scale Variational Mode Decomposition with Transfer Entropy for Real-Time Detection of Arrhythmia Transitions in Wearable ECG Devices," *Proc. 5th Int. Conf. on Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, 2026, pp. 1–6. doi:10.1109/ICECTE69292.2026.11429390. *Cited by 1*.

**[C3] Musavvir Haque**, M. Rana, S. Mustak, M. M. Hasan, and J. Islam, "FedSwarm-Attn: Federated Attention-Based Deep Reinforcement Learning for Energy-Efficient Multi-Robot Swarm Navigation in IoT Environments," *5th IEEE Int. Conf. on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)*, Jashore, Sep. 2026 (Presented; to appear in IEEE Xplore).

*Preprints*

**[P1]** N. M. Sathi, M. H. Rocky, N. R. Ritu, A. S. Fahim, **Musavvir Haque**, I. A. Mim, Md. Tariquzzaman, J. Islam, and Md. A. Hossain, "A Density Functional Theory Study of Penta-Boron Phosphide (P-$BP_5$) Nanosheet for NO, $SO_2$, $NO_2$, and $O_3$ Toxic Gas Sensing," SSRN preprint, posted 30 Jun. 2026. doi:10.2139/ssrn.7030770

## Research Experience

**Student Researcher**, ILAMPS, JUST — Jan 2025 – Present

- Student researcher at the Interdisciplinary Laboratory of Advanced Materials Processing and Simulation (ILAMPS), under Assoc. Prof. Md. Amzad Hossain, conducting MD/DFT-based computational modelling and ML-driven surrogate development for fusion-relevant materials and 2D nanosheet systems.
- Lead simulation-pipeline automation (Python/Bash) for high-throughput parallel HPC job management across multi-parameter MD/DFT sweeps.

## Research & Computational Skills

- **Atomistic Simulation:** LAMMPS (large-scale MD, NPT/NVT ensembles, cascade simulations, EAM/alloy potentials), ASE, pymatgen
- **DFT / Electronic Structure:** CASTEP (Materials Studio) – adsorption energies, DOS/PDOS, band-gap and charge-transfer analysis for 2D nanosheets; developing proficiency in VASP / Quantum ESPRESSO workflows
- **Machine Learning for Materials:** XGBoost, Random Forest, physics-informed feature engineering, multi-output surrogate modelling and regression pipelines (Python: scikit-learn, pandas, NumPy, Matplotlib, Seaborn)
- **Ion Implantation / Radiation Effects:** SRIM/TRIM simulations, implantation depth profiling, radiation damage metrics (DPA, NIEL)
- **Signal Processing:** Variational Mode Decomposition, Transfer Entropy, real-time wearable biosignal (ECG) processing, MATLAB
- **Engineering Tools:** SOLIDWORKS, Proteus, ModelSim, PVsyst, PLC (TIA Portal), PCB Design
- **Programming & Writing:** Python, Bash, MATLAB, Embedded C, Arduino; LaTeX, IEEE manuscript preparation, technical reporting

## Academic Leadership & IEEE Activities

**Chair**, IEEE Signal Processing Society JUST Student Branch Chapter — Sep 2025 – Dec 2025

- Led technical workshops and research mentorship in signal processing and computational sciences, and organised the *SigMa Mentorship Programme 2026* for undergraduate researchers.

**Vice Chair (Technical)**, IEEE JUST Student Branch — Apr 2024 – Sep 2025

- Organised seminars, workshops, and an international conference (ICETSD 2025) for EEE and CS students.

**Guest Speaker – Physical Design (PNR)**, EEE Club JUST — Jun 2025

- Invited seminar on Placement & Routing after industrial training at Ulkasemi Ltd., covering ASIC design, EDA tools, and semiconductor career paths.

**Speaker – Embedded AI / Speech & Audio Processing for Wearables** — Dec 2025

- Webinar on embedded AI, speech, and audio processing for wearable devices, organised by IEEE SPS DeKUT SBC with IEEE CAS & IEEE DeKUT SB.

**Judge – SignalQuest 2026 (IEEE SPS Day Quiz Competition)**, IEEE SPS JUST SBC — Jun 2026

## Honours & Awards

- **1st Place – Best Paper Presentation**, 2nd Int. Conf. on Engineering and Technology for Sustainable Development (ETSD 2026), JUST, May 2026 – "Integrated DFT–MD–ML Multiscale Modelling of Hydrogen Isotope Trapping and Oxide Evolution in Tungsten under Fusion-Relevant Plasma Exposure"
- **1st Place – Poster Presentation**, Sustainable IoT-Based Biometric Attendance System, JUST, May 2025

## Relevant Training & Certifications

- **IELTS Academic: Overall Band 7.0** (C1; Listening: 8.5, Reading: 7.0, Writing: 6.5, Speaking: 6.5), Oct 2026
- **Machine Learning With Python** – Bangladesh Computer Council (EDGE Project), JUST, Sep 2025 *(ID: EDGE-DSTS-115-6728-00001)*
- **Internship Trainee – Analog IC Design** (analog IC design, layout techniques, verification flows; VLSI, Cadence, ModelSim) – Ulkasemi Ltd., May 2025
- **Solar Training: PV Technology & System Design** – Daffodil International University, Jan 2026 *(ID: RE26312)*
- **Industrial Training – Electrical Engineering & Instrumentation** (PLC, sensors, automation), TICI, Jul 2025
- **Introduction to IoT** – Cisco, Jul 2023; **SOLIDWORKS Associate (CSWA) Course** – Udemy, Jul 2023

## Select Projects

**IoT-Based Dual-Mode Biometric Attendance System** (ESP32 + Supabase) Aug – Dec 2024

AES-256 encrypted, role-based cloud attendance system with offline sync; 1st place at JUST poster competition and accepted at IEEE QPAIN 2025.

**Energy Audit – Saleha Metal Industries Ltd.** (5.1M kWh/yr, 400 kVA transformer) May 2025

Industrial energy audit with recommendations for load-sharing optimisation and renewable energy integration.

## References

**Prof. Dr. Engr. Imran Khan**
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